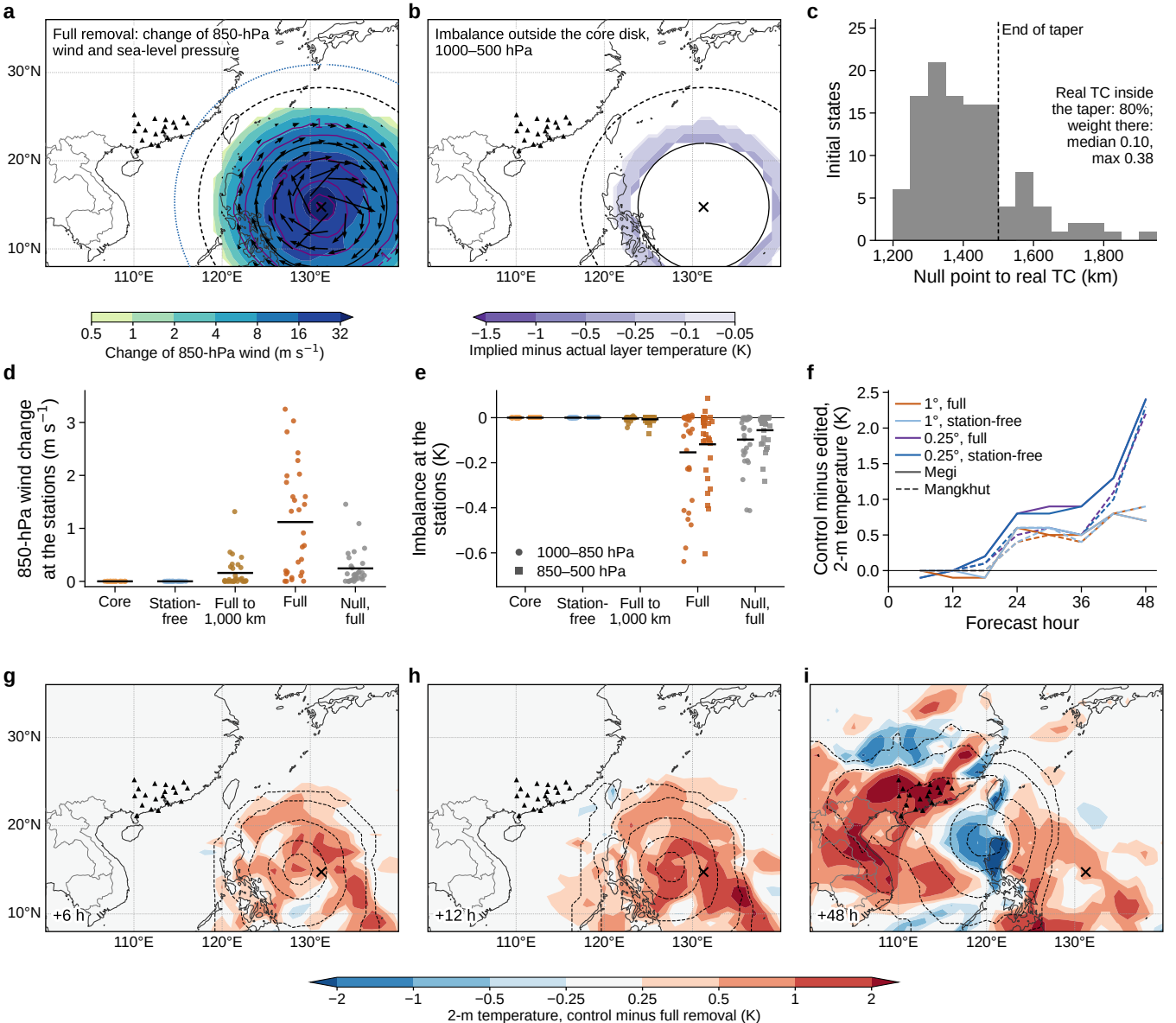

a
Full removal: change of 850-hPa wind and sea-level pressure
30°N
20°N
10°N
110°E
120°E
130°E
0.5
1
2
4
8
16
32
Change of 850-hPa wind (m s−1)
b
Imbalance outside the core disk, 1000–500 hPa
110°E
120°E
130°E
−1.5
−1
−0.5
−0.25
−0.1
−0.05
Implied minus actual layer temperature (K)
c
End of taper
Real TC inside the taper: 80%; weight there: median 0.10, max 0.38
Initial states
25
20
15
10
5
0
1,200
1,400
1,600
1,800
Null point to real TC (km)
d
850-hPa wind change at the stations (m s−1)
3
2
1
0
Core
Station-free
Full to 1,000 km
Full
Null, full
e
Imbalance at the stations (K)
0
−0.2
−0.4
−0.6
1000–850 hPa
850–500 hPa
Core
Station-free
Full to 1,000 km
Full
Null, full
f
Control minus edited, 2-m temperature (K)
2.5
2.0
1.5
1.0
0.5
0.0
1°, full
1°, station-free
0.25°, full
0.25°, station-free
Megi
Mangkhut
0
12
24
36
48
Forecast hour
g
30°N
20°N
10°N
+6 h
110°E
120°E
130°E
h
+12 h
110°E
120°E
130°E
i
+48 h
110°E
120°E
130°E
−2
−1
−0.5
−0.25
0.25
0.5
1
2
2-m temperature, control minus full removal (K)